Assignment 5: Assemble and Disassemble a PC Practical

Task1: List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Table 1: All Components inside a desktop PC after opening cabinet

Component	Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU	The brain of the computer that processes instructions and performs calculations.
CPU cooler	Keeps the CPU cool by transferring heat away from the processor and dissipating it into the air.
RAM	Temporary memory used to store data and programs currently being used by the CPU.
Storage Drive (SSD, HDD)	Stores the operating system, applications, and user files permanently. SSDs are faster than HDDs.
Power Supply Unit	Converts AC power from the wall into DC power for all internal computer components.
Graphics Card	Processes graphics and videos, providing smooth gaming, video editing, and display output.
Expansion Cards	Additional cards (e.g., Wi-Fi, sound card, capture card) installed to add extra functionality.
Case/Cabinet fans	Improve airflow inside the cabinet by bringing in cool air and exhausting hot air.
Optical Drive	Reads and writes CDs, DVDs, or Blu-ray discs. Many modern PCs do not include one.
CMOS Battery	Maintains BIOS/UEFI settings and keeps the system clock running when the PC is powered off.
SATA/Data Cables	Connect storage drives (HDD/SSD) to the motherboard for data transfer.
Power Cables	Deliver electrical power from the PSU to the motherboard, storage drives, GPU, and other components.
BIOS/UEFI Chips	Stores firmware that initializes the hardware and starts the operating system during boot.
Heat Sinks	Metal cooling blocks attached to components such as the chipset or SSD to dissipate heat.

Task 2: Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

Step 1: Shut Down and Unplug the power

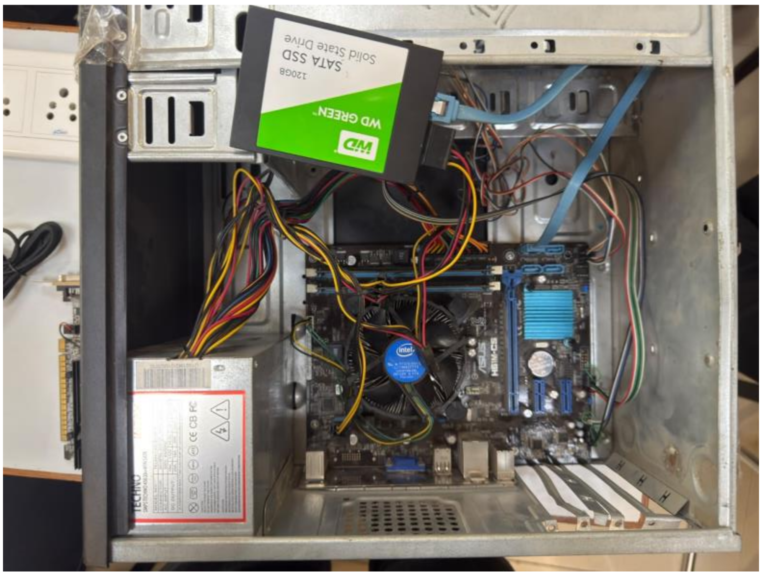
The desktop computer was completely shut down. The power cable and all external devices (keyboard, mouse, monitor, LAN cable, etc.) were unplugged to ensure safety before opening the cabinet.

Step 2: Remove the Side Panel

The screws securing the side panel were removed using a screwdriver. The side panel was then slid backward and lifted off to expose the internal components.

Step 3: Observe the Internal Components

The major internal components were identified, including the motherboard, CPU, RAM, power supply, storage drive, expansion slots, and cooling fan.

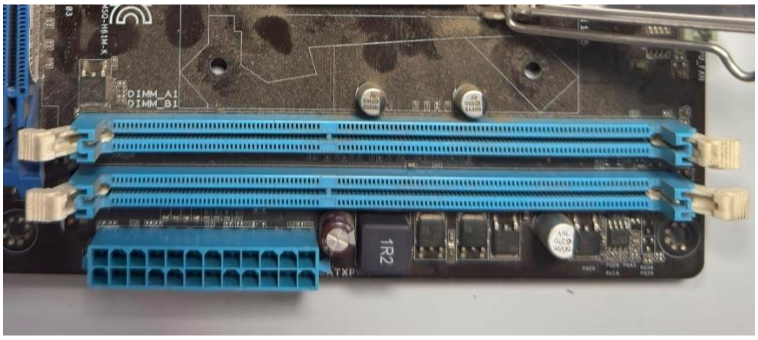


Step 4: Disconnect Power and Data Cables

The SATA data cables and power cables connected to the motherboard, hard drive, and other components were carefully disconnected before removing the hardware.

Step 5: Remove the RAM module

The locking clips on both sides of the RAM slot were pressed outward, and the RAM module was carefully lifted out while holding its edges.



Step 6: Remove the HDD/SSD

The SATA cable and power cable were disconnected. The screws holding the storage drive were removed, and the HDD/SSD was taken out of its mounting bay.

Step 7: Remove the Graphics Card

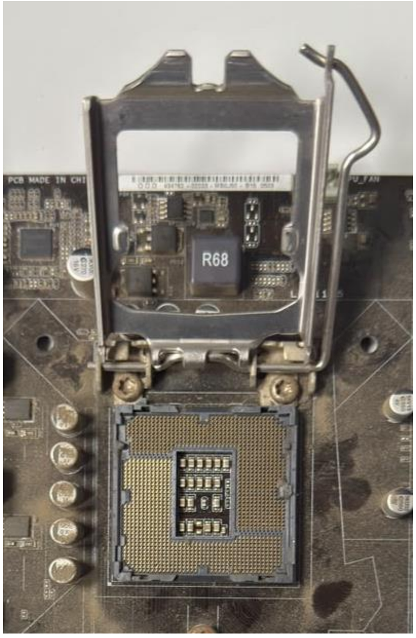
The PCIe locking clip was released, the retaining screw was removed, and the graphics card was carefully pulled out of the PCIe slot.

Step 8: Remove CPU Cooler

The CPU cooler's locking mechanism was released, and the fan cable was unplugged from the motherboard before removing the cooler.

Step 9: Remove the CPU

The CPU socket lever was lifted, and the processor was carefully removed by holding only its edges to avoid damaging the pins.



Step 10: Remove the Motherboard

After disconnecting all cables and removing the mounting screws, the motherboard was carefully lifted out of the computer cabinet.

Task 3: Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step 1: Install the Motherboard

The motherboard was carefully placed inside the cabinet, aligned with the standoffs, and secured using screws.

Step 2: Install the CPU

The CPU was aligned with the socket using the alignment triangle and gently placed into the socket. The socket lever was then locked.

Step 3: Install the CPU Cooler

The CPU cooler was mounted over the processor and locked in place. The CPU fan cable was connected to the **CPU_FAN** header on the motherboard.

Step 4: Install the RAM

The RAM module was aligned with the memory slot notch and pressed down until both locking clips clicked into place.

Step 5: Install the HDD/SSD

The storage drive was secured in its drive bay, and both the SATA data cable and SATA power cable were connected.

Step 6: Install the Graphics Card

The graphics card was inserted into the PCIe x16 slot and secured with a screw. PCIe power connectors were attached if required.

Step 7: Connect the Power Supply Cables

The 24-pin motherboard power connector, 4/8-pin CPU power connector, SATA power cables, and GPU power cables (if needed) were connected.

Step 8: Connect Front Panel and Data Cables

The front-panel connectors (Power SW, Reset SW, HDD LED, Power LED), USB headers, audio cable, and SATA data cables were connected to the motherboard.

Step 9: Organize Cables and Close the Cabinet

The cables were arranged neatly to improve airflow, and the side panel was reattached using screws.

Step 10: Connect External Devices and Power On

The monitor, keyboard, mouse, LAN cable, and power cable were connected. The computer was switched on to verify that all hardware was functioning correctly.

Task 4: Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Safety Precautions:

- Shut down the computer completely
- Unplug the power cable
- Disconnect all external devices
- Press the power button for 5-10 seconds after unplugging
- Wear an anti-static wrist strap or frequently touch a grounded metal object
- Work on a clean, flat, and non-conductive surface
- Handle components by their edges
- Avoid touching gold connectors and CPU pins
- Keep liquids and food away from the workspace
- Store screws and small parts in a container

Tools Required:

- Phillips-head screwdriver
- Anti-static mat
- Plastic cable ties
- Small container or magnetic tray
- Soft brush or compressed air
- Thermal paste
- Microfiber cloth